

ROYAL FITNESS CLUB

Body Transformation Blueprint

MODULE 4

Cardio for Body Composition

Duration	45 minutes
Course Code	CS_B3
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: The Role of Cardio in Body Transformation

Cardio: Tool, Not Treatment

Cardiovascular exercise is a tool — one of many in the body transformation toolkit. When used strategically, it accelerates fat loss, improves cardiovascular health, and supports recovery. When used excessively or inappropriately, it interferes with resistance training adaptations, increases recovery demands, and can drive muscle loss.

The key question is not "should I do cardio?" but "how much cardio, what type, and when — to maximise fat loss while preserving muscle?"

	Cardio Type	Duration	Intensity	Calorie Burn	Muscle Interference	Best For
LISS (Low-Intensity Steady State)	LISS (Low-Intensity Steady State)	30–60 min	50–65% HRmax	200–400 kcal/hr	Minimal	Weight loss; recovery; joint health; beginners
	MISS (Moderate)	20–40 min	65–75% HRmax	350–500 kcal/hr	Low	General conditioning; fat loss
HIIT (High-Intensity Interval Training)	HIIT (High-Intensity Interval Training)	15–25 min work/5–10 min rest	85–95% HRmax (300–600 kcal/hr + EPOC)	300–600 kcal/hr (+ EPOC)	Moderate	Fat loss; VO2 max; time efficiency
	Sprints (Anaerobic)	10–20 min	>95% HRmax (2000+ kcal/hr very short bursts)	2000+ kcal (very short bursts)	Moderation	Power, speed, fitness; body composition

SECTION 2: HIIT vs LISS — Evidence Review

What Actually Works Better for Fat Loss?

The HIIT vs LISS debate has generated hundreds of studies. The evidence shows:

Factor	HIIT Advantage	LISS Advantage
Time efficiency	Equal fat loss in 25 min vs 60 min	More steady state calorie burn per session if equal time available
EPOC	Significantly higher EPOC (12–24+ hours)	Minimal EPOC
Cardiovascular adaptation	Greater VO2 max gains	Mitochondrial density at low intensity
Muscle interference	Can interfere with strength/hypertrophy if excessive	Minimal interference even at high frequency
Recovery demand	High — needs 48h recovery	Low — can do daily without meaningful fatigue accumulation
Adherence	Lower (intensity makes it less enjoyable)	Higher (sustainable, pleasant, social)
Injury risk	Moderate-high (especially running-based)	Low

Evidence-based recommendation: Use primarily LISS (walking, cycling, swimming) for the bulk of cardio volume (3–5 sessions/week), adding 1–2 HIIT sessions/week for cardiovascular fitness and time efficiency. This preserves recovery capacity for resistance training while maximising fat-loss-supporting calorie expenditure.

SECTION 3: Concurrent Training — The Interference Effect

Can You Do Strength and Cardio Together?

The "interference effect" describes the well-documented phenomenon where concurrent endurance and resistance training produces inferior strength and hypertrophy gains compared to resistance training alone, and inferior endurance adaptations compared to endurance training alone.

Molecular mechanism: AMPK (activated by endurance training) inhibits mTORC1 (activated by resistance training). These two signalling pathways are essentially antagonistic — chronic concurrent training creates a chronic state of partial inhibition of both.

Mitigating the interference effect:

- Separate resistance and cardio sessions by at least 6 hours (ideally 24 hours)
- Perform resistance training first in any combined session; never cardio first
- Favour low-impact cardio (cycling, swimming) over running to reduce CNS and leg fatigue
- Limit HIIT to 2×/week maximum when hypertrophy is the primary goal
- Adequate caloric and protein intake is even more critical when doing concurrent training

Scenario	Interference Level	Strategy
Cardio before weights (same session)	HIGH	Always lift first; cardio depletes glycogen and creates pre-fatigue
Cardio immediately after weights	MODERATE	Acceptable; replenish carbs/protein first
Cardio 6h+ separate from weights	LOW	Good separation; AMPK/mTOR conflict minimised
Cardio on rest days from weights	VERY LOW	Ideal — no acute interference with lifting sessions
LISS cardio (walking) any time	MINIMAL	Walking AMPK stimulus is weak; virtually no interference

SECTION 4: Cardio Prescription for Transformation Phases

Periodising Cardio Around Your Training Goal

Phase	Goal	Resistance Training	Cardio Volume	Cardio Type
Muscle building (bulk)	Maximise hypertrophy	4-5 sessions/week, high volume	2-3 sessions/week, 20-30 mins	Low-impact, joint-decompressing (cycling, swimming)
Fat loss (cut)	Maximise fat loss while preserving muscle	3-4 sessions/week, lower volume	4-6 sessions/week, 30-60 mins	HIIT for VO2 maintenance
Maintenance/recomp	Maintain muscle, slow fat reduction	3-4 sessions/week	1-2 sessions/week, 30-45 mins	LISS and HIIT
Athletic performance	Sport-specific conditioning	2-3 sessions/week	1-2 sport-specific training sessions	System matched to sport demands

Module 4 (CS_B3) Key Takeaways

- Cardio is a tool to increase caloric expenditure; it does not selectively burn fat in isolation from caloric deficit
- LISS (walking, cycling) has minimal muscle interference and can be done 5–6×/week; HIIT: maximum 2×/week during hypertrophy phases
- The interference effect (AMPK vs mTOR conflict) is minimised by separating cardio from resistance training by 6+ hours
- Always lift weights before cardio in any combined session — never the reverse
- During a cut: increase cardio volume before decreasing calories — preserves energy for training and micronutrient adequacy
- Walking (LISS) is underrated: 10,000 steps/day burns 300–500 extra kcal with virtually zero recovery cost

1. Explain the AMPK-mTOR conflict (interference effect) at a molecular level.
2. Compare calorie burn of HIIT vs LISS including EPOC.
3. A client wants to do cardio immediately before their weight training session. What do you recommend and why?
4. How should cardio volume change when transitioning from a muscle-building phase to a fat loss phase?
5. Why is cycling a better cardio choice than running during a hypertrophy phase?